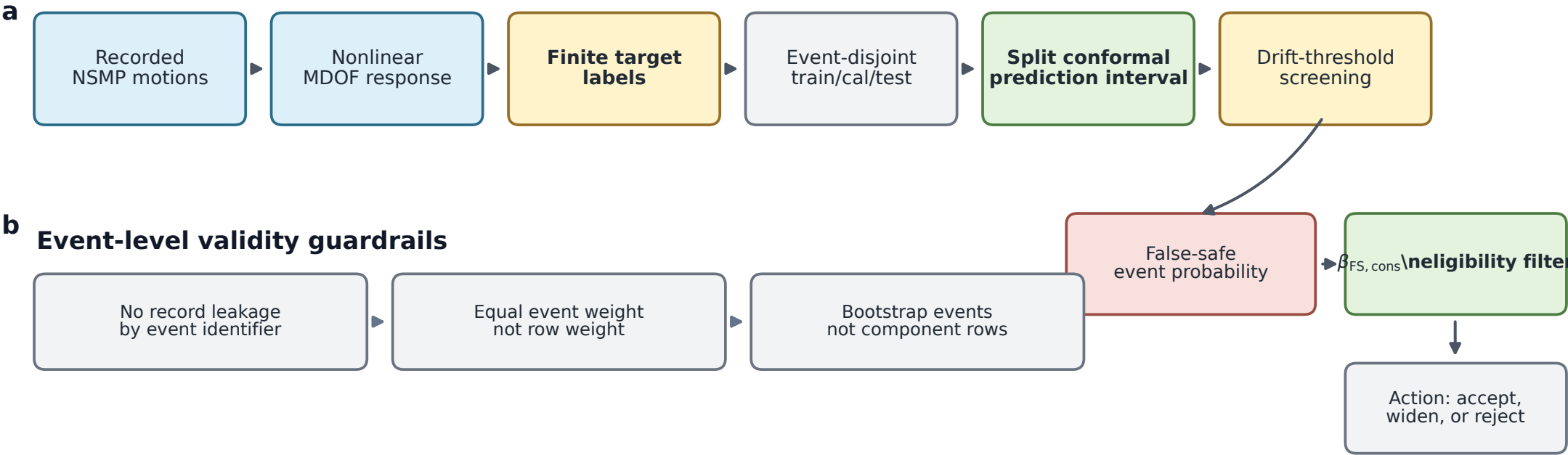


Research object and mechanism of reliability-aware seismic surrogate screening

Recorded motions drive nonlinear numerical response labels; event-level uncertainty controls a necessary false-safe eligibility filter.



c Interpretation for Structural Safety

The filter targets the critical error mode: unsafe numerical response classified as safe. Passing the filter is necessary, not sufficient, because excessive false-unsafe decisions can make a surrogate conservative but non-discriminatory for practical screening.

Response labels: numerical MDOF outputs

Ground motions: recorded public events